

LINUX



Keyboard Tricks Using [Tab]

All the Unix commands and the contents of the current directory are known to BASH (the Bourne Again SHell), so you don't need to type all the characters of a command or path. Just type in the first few and hit [Tab]—this will auto-complete the command for you. If you hear a beep, it doesn't mean an error: there's just more than one option with the same starting characters. Press [Tab] again to see a list of all the commands that begin with the characters you keyed in. The same applies to auto-completing a filename or path.

Say you want to switch from `/root` to `/usr/local/mystuff/videos`. This is how you use [Tab] in this case:

```
$cd /us [Tab]
```

auto-completes as: `/usr`. Now continue without pressing [Enter]:

```
$cd /usr/loc [Tab]
```

This auto-completes as: `/usr/local`.

Now, `$cd /usr/local/my [Tab]` auto-completes as `/usr/local/my\ stuff/`. Do not press [Enter] yet! We've deliberately included a space in the final folder name so as to demonstrate the importance or the usefulness of the [Tab] key when dealing with folder names containing spaces.

```
$cd /usr/local/my\ stuff/ [Tab]
```

As shown above, the space in a folder or file name is preceded with a backslash; it informs the shell of the space. Otherwise, the shell will consider it as two

separate files, "my" and "stuff."

If "my stuff" only contains one folder—as in our case, "videos"—then [Tab] will complete the path even if you don't type in any characters, since it is the obvious choice. The desired path using the auto-complete technique will appear as

```
$cd /usr/local/my\ stuff/videos
```

You can now press [Enter] to change to that directory. This technique can be applied to almost everything except with the "man" command.

Using [Ctrl] + [R]

Commands are sometimes called over and over again, perhaps with minor changes. Retyping them again and again even with the advantage of the auto-complete feature using [Tab] can be frustrating, especially if the command including its parameters is long. It also tests your memory! For example, the "cdrecord" command is very long, considering all the parameters involved. This is where the history of commands stored by BASH comes in handy.

To recall a command without having to remember and retype, press [Ctrl] + [R] and type the first few characters of the command. To see the various options you've used earlier with the command, press [Ctrl] + [R] over and over again until you find the command combination you're looking for. Pressing [Enter] will result in the execution of the chosen command. If any changes are to be made to the command, then hit [Tab] instead—this will display the command as if you just entered it.

As an example, when you press [Ctrl] + [R], the shell prompt changes to

```
(reverse-i-search)`':
```

Key in the first few characters of your command, say, "tar -". When you do this, BASH displays a few commands you had entered earlier, as in

```
(reverse-i-search)`tar`-:
```

```
tar -jxvf MPlayer-1.0pre7.tar.bz2
```

If the displayed command is not the one you're looking for, hit [Ctrl] + [R] again. It will display the next choice from the history of commands. Repeat this until it displays the one you are looking for. Say you're looking for "tar -C /usr/...". We shall repeat [Ctrl] + [R] until we get:

```
(reverse-i-search)`tar`-: tar -C /usr/more-apps/ -jxvf vlc-0.8.4a.tar.bz2
```

When the above line gets displayed, use [Tab] to bring it to the shell prompt, if any editing is to be made to the command; otherwise, hit [Enter] to directly execute the command.

Mount-Script

If you dual-boot Linux and Windows, you're likely to mount the Windows partition every now and then. Although some of the new distributions provide the option of mounting Windows partitions at startup, you might still want to keep control to yourself.

To have good control over the mounting of a partition, you need to be aware of the device name of a partition, such as `/dev/hda5` (a Windows FAT32 partition on our machine, also known as the E: drive) and a mount point for it (say `/mnt/win/E`). This is explained in the following.

To see the list of all the partitions on your hard disk, use the command "fdisk -l". Note down the device name of the your Windows drives; you need to mount these partitions to a predefined folder, called the "mount point." Create a folder called "win" under `/mnt`, and then create folders under "win" for the partitions you want to mount. We used the drive letters (C, D, and E) as the names for the mount points. When mounted, the Windows partitions would appear under `/mnt/win/C`, `/mnt/win/D`, and `/mnt/win/E`.

To mount your Windows D drive, say, you will have to issue the command

```
mount /dev/hdaX
```

`/mnt/win/D` where the "X" in "hdaX" stands for the partition number corresponding to the Windows D drive. You will have to repeat the command for all the partitions and at every boot. You'll soon find this a tedious task!

Simplify it: use just one command. Create shell scripts for mounting and un-mounting the partitions.

Assuming you're using konsole or gnome-terminal, create a file using the vi editor as follows:

```
#vi win-mount [Enter]
```

This creates a file with the name "win-mount", but it isn't saved yet. Press [Insert] to switch to the text editing mode of vi. Enter the following commands one below the other in the vi editor:

```
mount /dev/hdb1 /mnt/win/C
mount /dev/hdb5 /mnt/win/D
mount /dev/hdb6 /mnt/win/E
```

Save the file by first hitting [Esc] to exit from the typing/editing mode, then key in ":" (a colon) followed with "wq", which means save and exit.

Create another file in the same fashion to un-mount the mounted partitions. Only the file name (use "vi win-unmount") and the set of commands will undergo a few changes. The command will change to:

```
umount /mnt/win/C
umount /mnt/win/D
umount /mnt/win/E
```

Save and exit the file. The last step is to change this text file to a shell script. To achieve this, issue the following command:

```
#chmod 700 win-mount win-unmount [Enter]
```

Both the text files will now have shell script permissions. Whenever you want to mount the Windows partitions, change to the directory containing the script and issue

```
./win-mount [Enter]
```

To un-mount the partitions before shutdown or whenever you want to, change to the directory